

Shreya Komarabattini

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Education

Purdue University

Bachelor of Science in Computer Science

05/2026

Skills

Programming Languages: Python, Java, C, JavaScript, TypeScript

Web Technologies: HTML, CSS, React.js

Backend & Data: Node.js, Express, MongoDB, SQL

Tools & Platforms: Git, Jupyter Notebook, JavaFX, SceneBuilder

Testing: Cypress, Jest, Supertest

Data & AI: Data analysis, visualization, LLM, ChatGPT, Claude, Cursor, Prompt Engineering, Workflow Automation

Projects

ProtoPlay - Interactive Engineering Learning Platform (*Next.js, TypeScript, Tailwind CSS, shadcn/ui*)

LINK: <https://protoplay.vercel.app/>

01/2026

- Built an interactive educational platform that transforms computer science and system design concepts into visual simulations and hands-on learning experiences.
- Crafted 20+ challenge-based modules covering caching, scaling, load balancing, microservices, and distributed systems.
- Optimized rendering performance and user experience, achieving a 100 Lighthouse Performance score across core learning pages.

Healthycal - Mindful Eating Web Application (*React, Vite, Node.js, Express, mongodb, JWT, Cypress, Jest*)

LINK: <https://healthy-cal.vercel.app>

09/2025

- Enabled personalized dietary tracking for users by architecting a full-stack React/Node.js application with secure JWT authentication.
- Architected a backend using Express, MongoDB, and JWT authentication to securely manage user accounts, meal histories, and protected resources.
- Integrated nutrition search, meal analytics, and automated testing workflows with Cypress, Jest and Supertest to improve application stability.

IT Ticket Classification & Automation System (*Python, NLP, scikit-learn / transformers*)

LINK - <https://github.com/Shreyakb301/IT-Text-Classification.git>

8/2025

- Developed an NLP-based ticket classification system that automatically categorized IT support requests using TF-IDF vectorization and machine learning models.
- Built a FastAPI backend for real-time ticket prediction, automated prioritization, and workflow integration.

Experience

IT Lab Consultant, *Purdue University*

08/2023 – 05/2026

- Managed deployment, configuration, and maintenance of 100+ university lab computers, ensuring consistent software environments and security compliance.
- Resolved 150+ hardware and software incidents while maintaining 99% lab availability for students and faculty.
- Diagnosed and resolved system, printer, and network-related issues across multiple campus computing environments.
- Collaborated with university IT teams to support large-scale software rollouts, workstation upgrades, and campus technology initiatives across 7 campus buildings.

Undergraduate Researcher: Do We Really Know How to Use Graphs Effectively | Dr. Beomjin Kim

09/2024 – 02/2025

- Conducted research on how non-expert users interpret categorical data visualizations, including bar, line, and stacked bar charts.
- Designed and analyzed user studies to evaluate visualization effectiveness and identify common interpretation errors.
- Developed evidence-based recommendations for selecting chart types to improve communication of analytical insights.
- Presented findings at the 28th Annual Student Research and Creative Endeavors Symposium.

Teaching Assistant, Senior Capstone Project, *Purdue University*

02/2025 – 04/2025

- Assisted in evaluating capstone project reports and presentations, leveraging Python scripts to support data processing and review pipelines.

Summer Camp Assistant, *Purdue University*

06/2025 – 07/2025

- Represented the Computer Science department during a high-school summer camp program.
- Assisted students with technical activities and project development.
- Conducted research on prompt engineering and evaluated student capstone posters using AI-based tools.

Leadership & Presentations

- Active member of the Student Government Organization, organizing and supporting campus events.
- Presented research findings at the Love Data 2025 workshop at Purdue University, discussing data visualization practices and effective interpretation of categorical data.

